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Joyus AI — Brief for Robert Douglass

From: Alex UA, CEO, Zivtech **Date:** February 2026

What We're Building

Joyus AI is an open-source, multi-tenant AI agent platform that gives organizations mediated, skill-driven access to AI capabilities. The platform encodes organizational knowledge — business rules, compliance requirements, writing voice, quality standards — as testable, reusable skills that shape every AI interaction.

We're deploying to six organizations across healthcare (HIPAA), legal advocacy (attorney-client privilege, 30+ author voices), food manufacturing (FDA/USDA, cold-chain), a national museum, a large university, and the Law School Admissions Council. Each has radically different compliance requirements and operational realities.

How Spec Kitty Runs This Project

Spec Kitty is the development framework for every feature we build. Not an optional add-on — the entire project flows through it.

Five features managed through Spec Kitty so far:

Feature	Status	Work Packages	What It Does
001 MCP Server AWS Deployment	Completed	7 WPs	OAuth auth, Jira/Slack/GitHub/Google integrations, Docker deployment
002 Session Context Management	Accepted	9 WPs	Event-driven state snapshots, context handoff across session compaction
003 Platform Architecture Overview	Specified	—	Architecture decision records, layer definitions
004 Workflow Enforcement	Accepted	9 WPs	Quality gates, permission escalation, kanban state machine
005 Content Intelligence	Specified	—	129-feature stylometric attribution, writing profiles, fidelity monitoring

Each completed feature followed the full Spec Kitty lifecycle: constitution → specify → plan → tasks → implement (worktrees) → review → accept → merge. The workflow is enforced — you can't merge

without review, can't move tasks without committing spec changes first.

What we've built on top of Spec Kitty's workflow:

- **Worktree-per-work-package isolation.** Every WP gets its own git worktree. Implementation happens in isolation, merges back to main with `--no-ff`. Clean history, no cross-contamination between work packages.
- **Agent-compatible task management.** Claude Code operates as an implementing agent, using `spec-kitty agent tasks` commands to mark subtasks done, move work packages between lanes, and query status — all from within the AI session. The spec is the source of truth the agent works against.
- **Spec-driven quality gates.** The specification defines acceptance criteria before implementation starts. The review phase validates against those criteria. This is exactly the "specification as infrastructure" thesis that the whole platform is built on.

Where Spec Kitty Shines for AI-Driven Development

Three patterns emerged from building a full platform with Spec Kitty and Claude Code:

1. Specification prevents the biggest AI failure mode. AI agents are powerful but permissive — they'll build whatever you ask for, whether or not it's what you need. Spec Kitty forces the "what" to be defined before the "how." When Claude Code implements a work package, it has a spec to implement against, not a vague prompt. The difference in output quality is dramatic.

2. Work packages are the right unit of AI work. A full feature is too large for a single AI session — context windows fill up, sessions compact, state gets lost. A single task is too small to be meaningful. Work packages hit the sweet spot: scoped enough for one focused session, structured enough to produce a reviewable artifact. We routinely complete a WP per session with clean commits.

3. The kanban state machine creates accountability. When an AI agent moves a work package to `for_review`, there's a reviewable artifact with a clear spec to review against. The agent can't skip steps. The human reviewer has explicit acceptance criteria. This is the governance layer that makes AI-assisted development trustworthy.

What the Platform Does Beyond Development

Spec Kitty manages how we *build* the platform. The platform itself encodes organizational knowledge for *any* domain:

- **Six categories of skills:** Guardrails, operational context, report definitions, voice/brand, process logic, compliance — declared per tenant, enforced as hard failures.
- **Content Intelligence:** A 129-feature stylometric system validated at 97.9% attribution accuracy. Four fidelity tiers from quick writing skills (300 words) to forensic-grade profiles (50K+ words). Self-service profile building for any user.
- **Automated quality enforcement:** Pre-commit checks, coding standards, visual regression testing, accessibility audits — all configurable per project, enforced by the platform.
- **Automated pipelines:** Event-driven workflows (daily reports, regulatory change detection, bug triage) using the same skills and quality gates as human sessions.
- **Document & presentation generation, analysis tools, asset sharing** — all governed by the same skill system.
- **Model-agnostic, multi-tenant.** Claude is the initial backend, not the only one.

Where It Stands

- **Shipped:** MCP server, session context management, workflow enforcement, web chat UI
- **Fully specified:** Content Intelligence, profile engine, self-service profile building
- **Proven:** 97.9% authorship attribution, four fidelity tiers validated against EMNLP 2025 research
- **In roadmap:** Asset sharing, automated quality gates, presentation toolkit, document generator, analysis tools, visual regression testing, code execution sandbox
- **Deploying to:** Six organizations across healthcare, legal, food manufacturing, cultural, higher ed, and assessment
- **Open source:** Platform core public, Constitution v1.5 published

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The Conversation

Spec Kitty is foundational to how we work. I'd love to share what we've learned using it at scale on a real multi-feature platform — what works well, where we've built extensions, and where the tool could grow.

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